

**Question 5 : Solve any 3 linear equations of your choice using Gauss Elimination method.**

**1) FIRST EQUATION:**

$$\begin{aligned}x + y + z &= 2 \\x + 2y + 3z &= 5 \\x + 2y + 3z &= 5 \\2x + 3y + 4z &= 11\end{aligned}$$

STEP 1:

$$\begin{array}{ccc|c}1 & 1 & 1 & 2 \\1 & 2 & 3 & 5 \\2 & 3 & 4 & 11\end{array}$$

STEP 2:

$$\begin{array}{ccc|c}1 & 1 & 1 & 2 \\0 & 1 & 2 & 3 \\2 & 3 & 4 & 11\end{array}$$

STEP 3:

$$\begin{array}{ccc|c}1 & 1 & 1 & 2 \\0 & 1 & 2 & 3 \\0 & 1 & 2 & 7\end{array}$$

STEP 4:

$$\begin{array}{ccc|c}1 & 1 & 1 & 2 \\0 & 1 & 2 & 3 \\0 & 0 & 0 & 4\end{array}$$

STEP 4:

$$\begin{array}{ccc|c}1 & 1 & 1 & 2 \\0 & 1 & 2 & 3 \\0 & 0 & 0 & 1\end{array}$$

$$\begin{aligned}x - z &= -1 \\y + 2z &= 3 \\0 &= 4\end{aligned}$$

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**2)SECOND EQUATION:**

$$x + y + z = 9$$

$$2x + 5y + 7z = 52$$

$$2x + y - z = 0$$

STEP 1:

$$\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 2 & 5 & 7 & 52 \\ 2 & 1 & -1 & 0 \end{array}$$

STEP 2:

$$\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 2 & 5 & 7 & 52 \\ 0 & -1 & -3 & -18 \end{array}$$

STEP 3:

$$\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 2 & 5 & 7 & 52 \\ 0 & 0 & -12 & -70 \end{array}$$

STEP 4:

$$\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 2 & 5 & 7 & 52 \\ 0 & 0 & 1 & 35/3 \end{array}$$

STEP 5:

$$\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 2 & 5 & 0 & -13/3 \\ 0 & 0 & 1 & 35/3 \end{array}$$

STEP 6:

$$\begin{array}{ccc|c} 1 & 1 & 0 & -2/3 \\ 2 & 5 & 0 & -13/3 \end{array}$$

$$0 \ 0 \ 1 \mid 35/3$$

STEP 7:

$$\begin{array}{ccc|c} 1 & 1 & 0 & -2/3 \\ 0 & 4 & 0 & -11/3 \\ 0 & 0 & 1 & 35/3 \end{array}$$

STEP 8:

$$\begin{array}{ccc|c} 1 & 1 & 0 & -2/3 \\ 0 & 1 & 0 & -11/12 \\ 0 & 0 & 1 & 35/3 \end{array}$$

$$x = -2/3, y = -11/12, z = 35/3$$

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### 3) THIRD EQUATION:

$$4x - 5y = -6$$

$$2x - 2y = 1$$

STEP 1:

$$\begin{array}{cc|c} 4 & -5 & -6 \\ 2 & -2 & 1 \end{array}$$

STEP 2:

$$\begin{array}{cc|c} 4 & -5 & -6 \\ 0 & 1 & 4 \end{array}$$

STEP 3:

$$\begin{array}{cc|c} 4 & 0 & -26 \\ 0 & 1 & 4 \end{array}$$

$$y = 4$$

$$2x - 2(4) = 1 \quad 2x - 8 = 1 \quad 2x = 9 \quad x = 9/2$$

$$x = 9/2 \text{ and } y = 4.$$

